SYSTEM PROPOSAL

WeatherWise : Real-Time Weather Monitoring and Forecasting

Platform: Web Application

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1. Introduction

Weather can change throughout the day, so having current information can help people plan their activities and travel. Checking several sources can also be inconvenient when users only need basic weather details for a specific location.

WeatherWise provides a simple place to search for a location and view its current weather information. The system focuses on the details users commonly check, including temperature, weather condition, humidity, wind speed, and rain forecast.

The proposed Web application will use a weather API to retrieve updated weather data and will be designed to work on different screen sizes so users can check the information using phones, tablets, or computers.

2. Purpose of the Project

This proposal aims to present a Web application that will address the difficulty of quickly checking weather information for different locations, streamline weather monitoring, and improve access to current conditions and forecasts.

3. Objectives

3.1 General Objective

To design and develop a responsive web-based weather monitoring system that retrieves current weather information and forecasts for user-selected locations.

3.2 Specific Objectives

- To provide users with a location search for checking weather information.

- To display current temperature and weather conditions for the selected location.
- To display humidity and wind speed as part of the current weather information.
- To provide a rain forecast and upcoming weather forecast.
- To integrate a weather API for retrieving updated weather data.
- To provide a responsive interface that can be used on different screen sizes.

4. Project Scope

4.1 In Scope

- Location search.
- Current weather information.
- Temperature Display.
- Weather condition display.
- Humidity and wind speed details.
- Rain forecast and upcoming forecast.
- Weather API integration.
- Responsive Web design.

4.2 Out of Scope

- Weather hardware or physical weather sensors.
- Professional meteorological forecasting.
- Emergency warning or disaster alert services.
- Weather data collection without an external weather API.
- Native mobile applications.
- Paid weather subscriptions or premium forecasting services.

5. System Features / Functions

Module	Feature	Description
Weather Search	Location Search	Allows users to search for a location before viewing its weather information.
Current Weather	Weather Details	Displays temperature, weather conditions, humidity, and wind speed.
Forecast	Rain Forecast	Shows rain-related forecast information for the selected location.
Forecast	Upcoming Forecast	Displays upcoming weather information for the selected location.
API Integration	Weather API	Retrieves updated current weather and forecast data.
Interface	Responsive Design	Adjusts the web interface for phones, tablets, and desktop screens.

Module	Feature	Description
Administration	System Settings	Allows the administrator to manage basic system settings.

6. Technology Stack

Layer	Technology / Tool	Purpose
Frontend (Web)	React, Vite, React Router	Builds the responsive interface for location search, current weather, and forecasts.
Backend / API	Node.js, Express	Handles API requests and server-side processing.
Database	MySQL	Can store user preferences or saved locations if these are included in development.
External API	OpenWeather API	Provides current weather information and forecast.
Cloud / Hosting	Render / Vertical (proposed)	Deployment options for the Web applications.
Version Control	Git / Github (proposed)	Stores and tracks source-code changes.
Project Management Tool	Trello / Notion (proposed)	Task tracking and project progress.

7. Target Users

User Role	Description	Key Needs / Usage
User	Person checking weather information.	Searches locations and views current conditions and forecasts.
Admin	Manages basic system settings and access if administrative functions are included.	Maintains system-related settings.

8. Project Timeline / Work Plan

Phase	Activities	Estimated Duration
Planning & Requirements Gathering	Review the project workflow, users, requirements, and main features.	1 week
System Design	Plan the interface, API flow, location search, and responsive layout.	2 weeks
Development	Build the frontend, backend, API integration, and weather display functions.	4 weeks

Phase	Activities	Estimated Duration
Testing & QA	Test location search, weather data, forecasts, API responses, and responsive pages.	2 weeks
Deployment	Configure the database. environment variables, hosting, and project documentation.	1 week
Project Presentation	Presentation of project	Ongoing / Final